

POWER BI ASSIGNMENT 1 REPORT

Module -2

Data Transformation & Data Modeling

Project Title

E-Commerce Sales Data Analysis using Power BI

SUBMITTED BY

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Introduction

This assignment focuses on performing data preparation, transformation, and data Modeling using Power BI. An E-commerce dataset consisting of multiple related tables was used to clean, transform, and integrate data. The objective was to ensure data consistency, apply business logic, and establish relationships between datasets for effective analysis.

Dataset Description

The following datasets were used in this assignment:

1. **List of Orders** – (Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit)
2. **Order Details** - (Order ID, Category, Sub-Category, Quantity, Amount, Profit)
3. **Sales Target** – (Category, Month of Order Date, Target)

1. Data Import & Transformation

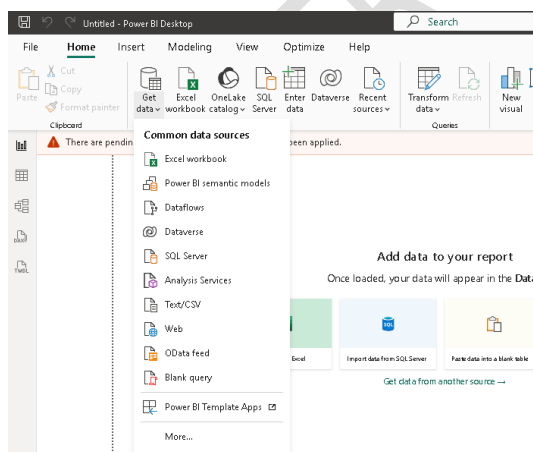
Task:

- Import *List of Orders.csv* into Power BI.
- Open *the list of orders* in the Power Query Editor using the Transform Data option.
- Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.

Answer:

All three datasets were imported into Power BI using the “Get Data” & Selected Text/CSV(since the dataset is CSV format file) option and loaded into the Power Query Editor for transformation.

2.Row Limiting & Data Types:



List of Orders.csv

Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonalshi	Jammu and Kashmir	Srinagar
B-25608	08-04-2018	Amarshi	Tamil Nadu	Chennai
B-25609	09-04-2018	Heash	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Reneek	Madhya Pradesh	Bhopal

Task: Restrict the List of Orders table to the first 500 rows.

Answer: The List of Orders table was limited to the first 500 rows. After applying Use First Row as Headers, the Keep Top Rows transformation was used to retain only 500 records.

Queries [3] fx = Table.PromoteHeaders(#"Change Type", [PromoteAllScalars=true])

	A ^B Order ID	A ^B Order Date	A ^B CustomerName	A ^B State	A ^B City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna

Query Settings

PROPERTIES
Name
List of Orders

APPLIED STEPS
Source
Change Type
Changed First row as headers
Changed Type

fx = Table.FirstN(#"Changed Type",500)

	A ^B Order ID	A ^B Order Date	A ^B CustomerName	A ^B State	A ^B City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram

Query Settings

PROPERTIES
Name
List of Orders

APPLIED STEPS
Source
Change Type
Changed First row as headers
Changed Type
Kept First 500 Rows

Task: Set the Order Date column to the Date data type.

Answer The data types were initially updated from Text to Date for the *Order Date* column. However, errors were encountered due to inconsistent date formats within the dataset. The "Change Type" step was deleted and the *Order Date* column was converted using "Change Type with Locale" to correctly handle the date format and avoid errors.

fx = Table.TransformColumnTypes(#"Removed Duplicates",{"Order Date", type date})

	A ^B Order ID	Order Date	A ^B CustomerName	A ^B State	A ^B City
1	B-25601	1/4/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	1/4/2018	Pearl	Maharashtra	Pune
3	B-25603	3/4/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	3/4/2018	Divsha	Rajasthan	Jaipur
5	B-25605	5/4/2018	Kasheen	West Bengal	Kolkata
6	B-25606	6/4/2018	Hazel	Karnataka	Bangalore
7	B-25607	6/4/2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	8/4/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	9/4/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	9/4/2018	Yogesh	Bihar	Patna
11	B-25611	11/4/2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12/4/2018	Shrichand	Punjab	Chandigarh
13	B-25613	12/4/2018	Mukesh	Haryana	Chandigarh
14	B-25614	Error	Vandana	Himachal Pradesh	Simla
15	B-25615	Error	Bhavna	Sikkim	Gangtok
16	B-25616	Error	Kanak	Goa	Goa
17	B-25617	Error	Sagar	Nagaland	Kohima
18	B-25618	Error	Manju	Andhra Pradesh	Hyderabad
19	B-25619	Error	Ramesh	Gujarat	Ahmedabad
20	B-25620	Error	Sarita	Maharashtra	Pune
21	B-25621	Error	Deepak	Madhya Pradesh	Bhopal
22	B-25622	Error	Monisha	Rajasthan	Jaipur
23	B-25623	Error	Atharv	West Bengal	Kolkata

APPLIED STEPS
Source
Change Type
Promoted Headers
Changed Type
Kept First Rows
Removed Duplicates
Changed Type1

fx = Table.TransformColumnTypes(#"Kept First 500 Rows", {"Order Date", type date}, "en-IN")

	A ^B Order ID	Order Date	A ^B CustomerName	A ^B State	A ^B City
1	B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	4/1/2018	Pearl	Maharashtra	Pune
3	B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	4/3/2018	Divsha	Rajasthan	Jaipur
5	B-25605	4/5/2018	Kasheen	West Bengal	Kolkata
6	B-25606	4/6/2018	Hazel	Karnataka	Bangalore
7	B-25607	4/6/2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	4/9/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	4/9/2018	Yogesh	Bihar	Patna
11	B-25611	4/11/2018	Anita	Kerala	Thiruvananthapuram

Query Settings

PROPERTIES
Name
List of Orders

APPLIED STEPS
Source
Change Type
Changed First row as headers
Changed Type
Kept First 500 Rows
Changed Data Type using Locale

Task: Change the Amount and Target columns to fixed decimal numbers.

Result: Data types were updated:

- Order Date → Date format
- Amount & Target → Fixed Decimal Number

Query Settings: Table.TransformColumnTypes(#"Change Type",{"Amount", Currency.Type})

Order ID	Amount	Profit	Quantity	Category
1	1,275.00	-148	7	Furniture
2	66.00	-12	5	Clothing
3	8.00	-2	3	Clothing
4	80.00	-56	4	Electronics
5	168.00	-111	2	Electronics
6	424.00	-272	5	Electronics
7	2,617.00	1151	4	Electronics
8	561.00	212	3	Clothing
9	119.00	-5	8	Clothing
10	1,355.00	-60	5	Clothing

APPLIED STEPS: Source, Use First Row as Headers, Change Type, Changed Amount to Fixed Decimal

Query Settings: Table.TransformColumnTypes(#"Change Type",{"Target", Currency.Type})

Month of Order Date	Category	Target
1	4/18/2016 Furniture	10,400.00
2	5/18/2016 Furniture	10,500.00
3	6/18/2016 Furniture	10,600.00
4	7/18/2016 Furniture	10,800.00
5	8/18/2016 Furniture	10,900.00
6	9/18/2016 Furniture	11,000.00
7	10/18/2016 Furniture	11,100.00
8	11/18/2016 Furniture	11,300.00
9	12/18/2016 Furniture	11,400.00
10	1/19/2016 Furniture	11,500.00

APPLIED STEPS: Source, Use First Row as Headers, Change Type, Changed Target to Fixed Decimal Nu...

3.Text Formatting

- The Customer Name column was formatted to Proper Case to ensure consistent capitalization.
- Transform-> Format -> Capitalize each word

Query Settings: Table.TransformColumns(#"Changed Data Type using Locale",{"CustomerName", Text.Proper, type text})

Order ID	Order Date	CustomerName	State	City
1	4/1/2018	Bharat	Gujarat	Ahmedabad
2	4/1/2018	Pearl	Maharashtra	Pune
3	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4	4/3/2018	Divsha	Rajasthan	Jaipur
5	4/5/2018	Kasheen	West Bengal	Kolkata
6	4/6/2018	Hazel	Karnataka	Bangalore
7	4/6/2018	Sonakshi	Jammu and Kashmir	Kashmir
8	4/8/2018	Aarushi	Tamil Nadu	Chennai
9	4/9/2018	Jitesh	Uttar Pradesh	Lucknow
10	4/9/2018	Yogesh	Bihar	Patna
11	4/11/2018	Anita	Kerala	Thiruvananthapuram
12	4/12/2018	Shrichand	Punjab	Chandigarh
13	4/12/2018	Mukesh	Haryana	Chandigarh

APPLIED STEPS: Source, Change Type, Changed First row as headers, Changed Type, Kept First 500 Rows, Changed Data Type using Locale, Changed Customer Name to Proper ...

4.Column Creation

- Merge the **City** and **State** columns to create a new column named **Location** in the format: **City, State**

Query Settings: Table.CombineColumns(#"Changed Customer Name to Proper Case",{"State", "City"},

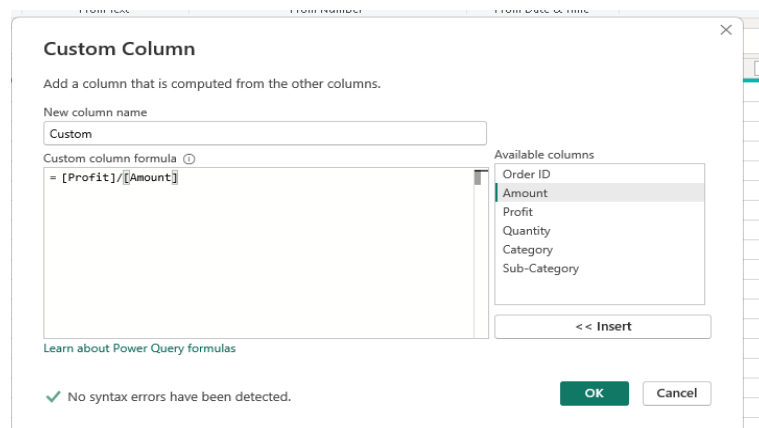
Order ID	Order Date	CustomerName	Location
1	4/1/2018	Bharat	Gujarat, Ahmedabad
2	4/1/2018	Pearl	Maharashtra, Pune
3	4/3/2018	Jahan	Madhya Pradesh, Bhopal
4	4/3/2018	Divsha	Rajasthan, Jaipur
5	4/5/2018	Kasheen	West Bengal, Kolkata
6	4/6/2018	Hazel	Karnataka, Bangalore
7	4/6/2018	Sonakshi	Jammu and Kashmir, Kashmir
8	4/8/2018	Aarushi	Tamil Nadu, Chennai
9	4/9/2018	Jitesh	Uttar Pradesh, Lucknow
10	4/9/2018	Yogesh	Bihar, Patna
11	4/11/2018	Anita	Kerala, Thiruvananthapuram
12	4/12/2018	Shrichand	Punjab, Chandigarh
13	4/12/2018	Mukesh	Haryana, Chandigarh

APPLIED STEPS: Source, Change Type, Changed First row as headers, Changed Type, Kept First 500 Rows, Changed Data Type using Locale, Changed Customer Name to Proper ...

- Create a custom column named **Profit Margin** calculated as: **Profit / Amount** (expressed as a percentage)

Two new columns were created:

- **Location Column:** Combined City and State in the format *City, State*
- **Profit Margin Column:** Calculated using formula:
Profit Margin = Profit / Amount



= Table.ReorderColumns("#Created Profit Margin using 2 columns",{"Order ID", "Amount", "Profit", "Profit Margin", "Quantity"})						
	Order ID	Amount	Profit	Profit Margin	Quantity	
1	5601	1,275.00	-1148	-0.900392157	7	
2	5601	66.00	-12	-0.181818182	5	
3	5601	8.00	-2	-0.25	3	
4	5601	80.00	-56	-0.7	4	
5	5602	168.00	-111	-0.660714286	2	
6	5602	424.00	-272	-0.641509434	5	
7	5602	2,617.00	1151	0.439816584	4	
8	5602	561.00	212	0.377896613	3	
9	5602	119.00	-5	-0.042016807	8	
10	5603	1,355.00	-60	-0.044280443	5	
11	5603	24.00	-30	-1.25	1	
12	5603	193.00	-166	-0.860103627	3	
13	5603	120.00	0	0.000000000	2	

5. Conditional Column

Add a conditional column named **Profit Status** based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

Answer: Created a Conditional Column based on the given conditions.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Profit Status

	Column Name	Operator	Value ①		Output ①
If	Profit	is less than	0	Then	Loss
Else If	Profit	equals	0	Then	Break-Even
Else If	Profit	is greater than	0	Then	Profit

Add Clause

Else ①

ABC 123 null

OK

Cancel

✖

✓

fx

= Table.AddColumn("#Reordered Profit Margin column near Profit Column", "Profit Status", each if

ABC 123

Profit Margin

123

Quantity

ABC C

Category

ABC 123

Sub-Category

ABC 123

Profit Status

1

-0.900392157

7

Furniture

Bookcases

Loss

2

-0.181818182

5

Clothing

Stole

Loss

3

-0.25

3

Clothing

Hankerchief

Loss

4

-0.7

4

Electronics

Electronic Games

Loss

5

-0.660714286

2

Electronics

Phones

Loss

6

-0.641509434

5

Electronics

Phones

Loss

7

0.439816584

4

Electronics

Phones

Profit

8

0.377896613

3

Clothing

Saree

Profit

9

-0.042016807

8

Clothing

Saree

Loss

10

-0.044280443

5

Clothing

Trousers

Loss

11

-1.25

1

Furniture

Chairs

Loss

12

-0.860103627

3

Clothing

Saree

Loss

13

0.027777778

3

Clothing

Trousers

Profit

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Use First Row as Headers

Change Type

Changed Amount to Fixed Decimal

Created Profit Margin using 2 columns

Reordered Profit Margin column near...

Created Profit Status Column

6.Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.

Answer:

- The List of Orders and Order Details tables were merged using the **Order ID** column.
- The resulting table was named **Orders Data**.

Merge

Select a table and matching columns to create a merged table.

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25601	66.00	-12	5	Clothing	Stole	-18.18%	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00%	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss

List of Orders

Order ID	Order Date	CustomerName	Location
B-25601	4/1/2018	Bharat	Ahmedabad, Gujarat
B-25602	4/1/2018	Pearl	Pune, Maharashtra
B-25603	4/3/2018	Jahan	Bhopal, Madhya Pradesh
B-25604	4/3/2018	Divsha	Jaipur, Rajasthan
B-25605	4/5/2018	Kasheen	Kolkata, West Bengal

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

= Table.ExpandTableColumn("#Merged 2 tables and created Orders Data", "Order Details", {"Amount",

Profit	Profit Margin	Quantity	Category	Sub-Category
-1148	-0.900392157	7	Furniture	Bookcases
-12	-0.181818182	5	Clothing	Stole
-2	-0.25	3	Clothing	Hankerchief
-56	-0.7	4	Electronics	Electronic Games
-111	-0.660714286	2	Electronics	Phones
-272	-0.641509434	5	Electronics	Phones
1151	0.439816584	4	Electronics	Phones
212	0.377896613	3	Clothing	Saree
-5	-0.042016807	8	Clothing	Saree
-60	-0.044280443	5	Clothing	Trousers
-30	-1.25	1	Furniture	Chairs
-166	-0.860103627	3	Clothing	Saree
5	0.027777778	3	Clothing	Trousers
16	0.137931034	4	Clothing	Stole
36	0.336448598	6	Clothing	Stole

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Change Type

Changed First row as headers

Changed Type

Kept First 500 Rows

Changed Data Type using Locale

Changed Customer Name to Proper ...

Created a New Column named Locati...

Merged 2 tables and created Orders ...

Expanded Order Details

7. Handling Missing Data & Duplicate Data

- All datasets were checked for missing values.

= Table.Distinct("#Expanded Order Details")

Order ID	Order Date	CustomerName
1 B-25601		
2 B-25601		
3 B-25601		
4 B-25601		
5 B-25602		
6 B-25602		
7 B-25602	4/1/2018	Pearl

Order Date

1000 (100%) Valid

0 (0%) Error

0 (0%) Empty

fx = Table.Distinct("#Expanded Order Details")

Order ID	Order Date	CustomerName	Location	\$ Amount
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1 B-25601	4/1/2018	Bharat	Gujarat, Ahmedabad	1,2

- Missing values were identified using Column Quality.
- Null values in numeric columns such as Profit and Quantity were replaced with 0.
- Rows with missing Order ID were removed as they are critical fields. Some columns were retained with null values where appropriate.

8.Sorting & Filtering

- Sort records in the Orders Data table by Order Date in descending order.
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Answer:

- Data was sorted by **Order Date in descending order**.

fx = Table.Sort("#Removed Blank Rows",{{"Order Date", Order.Descending}})

Order ID	Order Date	CustomerName	Location	\$ Amount
B-26100	3/31/2019	Hitika	Madhya Pradesh, Indore	
B-26100	3/31/2019	Hitika	Madhya Pradesh, Indore	
B-26100	3/31/2019	Hitika	Madhya Pradesh, Indore	
B-26099	3/30/2019	Bhishm	Maharashtra, Mumbai	
B-26099	3/30/2019	Bhishm	Maharashtra, Mumbai	2,5
B-26099	3/30/2019	Bhishm	Maharashtra, Mumbai	
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	4
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	4
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	
B-26098	3/29/2019	Pinky	Jammu and Kashmir, Kashmir	
B-26096	3/28/2019	Atharv	West Bengal, Kolkata	1
B-26097	3/28/2019	Vini	Karnataka, Bangalore	
B-26097	3/28/2019	Vini	Karnataka, Bangalore	
B-26096	3/28/2019	Atharv	West Bengal, Kolkata	
B-26097	3/28/2019	Vini	Karnataka, Bangalore	

Query Settings

PROPERTIES

Name
List of Orders

APPLIED STEPS

- Source *
- Change Type *
- Changed First row as headers *
- Changed Type *
- Kept First 500 Rows *
- Changed Data Type using Locale *
- Changed Customer Name to Proper ... *
- Created a New Column named Locati... *
- Merged 2 tables and created Orders ... *
- Expanded Order Details *
- Removed Duplicates *
- Replaced Null *
- Removed Blank Rows *
- Order Date sorted by Descending

- Filters were applied to analyse data for specific states (e.g., Tamil Nadu).

= Table.SelectRows(#"Order Date sorted by Descending", each ([Location] = "Tamil Nadu, Chennai"))					Query Settings
Order Date	CustomerName	Location		Amount	PROPERTIES
					Name
					List of Orders
					All Properties
					APPLIED STEPS
					Source
					Change Type
					Changed First row as headers
					Changed Type
					Kept First 500 Rows
					Changed Data Type using Locale
					Changed Customer Name to Proper ...
					Created a New Column named Locati...
					Merged 2 tables and created Orders ...
					Expanded Order Details
					Removed Duplicates
					Replaced Null
					Removed Blank Rows
					Order Date sorted by Descending
					Filtered Tamilnadu Location

9.Grouping and Aggregating Data

Duplicate the *Order Details* table and perform the following:

- Count of Order ID
- Average Profit by Category
- Total Amount by Sub-Category

Answer: A duplicate of the Order Details table was created. The Basic Group By option was first used, but since it supports only one aggregation and removes other columns, the Advanced option was used to perform multiple calculations.

Group By

Specify the columns to group by and one or more outputs.

☐ Basic ☒ Advanced

Order ID
Category
Sub-Category

Add grouping

New column name	Operation	Column
Count	Count Rows	
Average Profit	Average	Profit
Total Amount	Sum	Amount

Add aggregation

OK Cancel

Task: Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date**.

Answer:

- Sales Target table was duplicated.
- Total Target was calculated by Month of Order Date.

= Table.Group("#Removed Duplicates", {"Month of Order Date"}, {"Total Target", each List.Sum(		
Month of Order Date	1.2 Total Target	
1	4/18/2026	31400
2	5/18/2026	31500
3	6/18/2026	31600
4	7/18/2026	33800
5	8/18/2026	33900
6	9/18/2026	34000
7	10/18/2026	36100
8	11/18/2026	36300
9	12/18/2026	36400
10	1/19/2026	43500
11	2/19/2026	43600
12	3/19/2026	43800

Query Settings

PROPERTIES

Name: Sales target_Duplicate

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Changed Target to Fixed Decimal Nu...
- Removed Duplicates
- Grouped Target using Order Date

10.Data Modeling:

Task:

- Establish a relationship between:
 - List of Orders and Order Details using Order ID
- Establish a relationship between:
 - Order Details and Sales Target using Category

Answer:

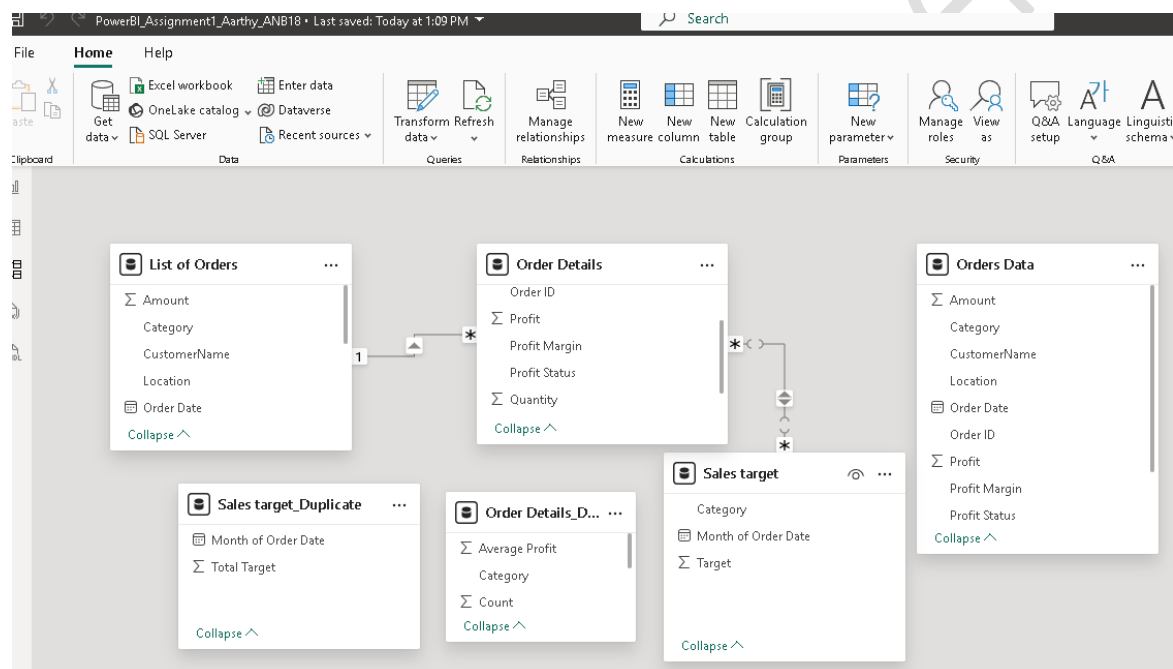
Step1: Given Close & Apply

Step 2: From Left Panel, Clicked Model View

Relationships were established between tables:

Relationships Created:

- List of Orders ↔ Order Details (using Order ID)
- Order Details ↔ Sales Target (using Category)
- Relationships were verified using the **Manage Relationships** option and ensured to be active.



Manage relationships

+ New relationship		Autodetect	Edit	Delete	Filter
☐ From: table (column) ↑	Relationship	To: table (column)	Status		
☐ Order Details (Category)		Sales target (Category)	Active	...	
☐ Order Details (Order ID)		List of Orders (Order ID)	Active	...	

Conclusion

The dataset was successfully cleaned, transformed, and structured using Power Query. Necessary business logic was applied through calculated and conditional columns. Data from multiple tables was integrated using joins, and relationships were established to create a reliable data model. This prepared dataset is now suitable for further analysis and visualization in Power BI.

POWER BI Assignment 1